Motion of particles on a friction surface

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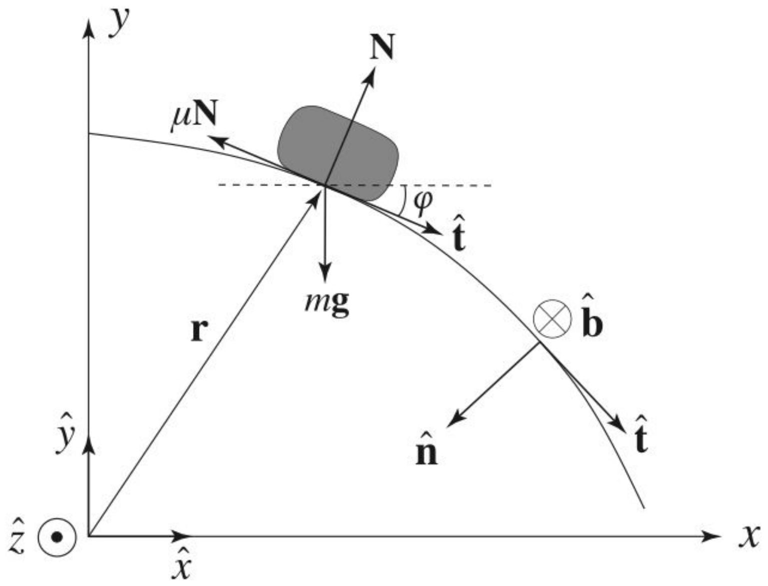
Introduction & Contextual Background

Introduction

The motion of an object sliding on a curved surface is a classical mechanics problem involving gravity, friction, and surface geometry. This study is based on *Motion of a Block Slipping on an Arbitrary Surface with Friction* (Gonzalez-Cataldo et al., 2017), which presents a general framework for predicting when an object loses contact with a curved surface. The model relates the particle's motion to the local curvature of the surface and introduces a horizon function to determine the departure point. In this project, the theory was applied to a hemispherical acrylic surface and compared with experimental measurements.

Kinematic



$$\mathbf{r}(t) = (x(t), y(t))$$

$$s(t) = \int_{t_0}^{t} \left\| \frac{d\mathbf{r}(\tau)}{d\tau} \right\| d\tau, \quad \text{where } t \in [t_0, T],$$

$$\hat{\mathbf{t}} \equiv \frac{d\mathbf{r}}{ds},$$

$$\hat{\mathbf{n}} \equiv \frac{1}{\kappa(s)} \frac{d\hat{\mathbf{t}}}{ds},$$

$$\hat{\mathbf{b}} \equiv \hat{\mathbf{t}} \times \hat{\mathbf{n}},$$

Kinematic

$$\mathbf{v}(s) = v(s)\hat{\mathbf{t}}(s)$$

$$\mathbf{a}(s) = v(s)\frac{dv(s)}{ds}\hat{\mathbf{t}}(s) + v^2(s)\kappa(s)\hat{\mathbf{n}}(s).$$

- Velocity is always directed along the tangent vector.
- The first term represents the tangential acceleration (change in speed).
- The second term represents the normal acceleration (change in direction).

Dynamics in the Frenet-Serret frame

<the tangential equation of motion>

$$mv(s)\frac{dv(s)}{ds} = m\mathbf{g}\cdot\hat{\mathbf{t}}(s) - \mu N(s)$$

<the normal-direction equation of motion>

$$m\kappa(s)v^2(s) = m\mathbf{g}\cdot\hat{\mathbf{n}}(s) + (\hat{\mathbf{b}}\cdot\hat{\mathbf{z}})N(s).$$

<the most general equation representing the velocity of a particle moving on a curved surface>

$$v^2(s) = v_0^2\, e^{-\left(2\mu/\hat{\mathbf{b}}\cdot\hat{\mathbf{z}}\right)\Phi(s)} + 2\, e^{-\left(2\mu/\hat{\mathbf{b}}\cdot\hat{\mathbf{z}}\right)\Phi(s)} \times \int_0^s \left[\mathbf{g}\cdot\hat{\mathbf{t}}(s') + \mu\frac{\mathbf{g}\cdot\hat{\mathbf{n}}(s')}{\hat{\mathbf{b}}\cdot\hat{\mathbf{z}}}\right] e^{\left(2\mu/\hat{\mathbf{b}}\cdot\hat{\mathbf{z}}\right)\Phi(s')} ds$$

Tangent angle

Eq.1

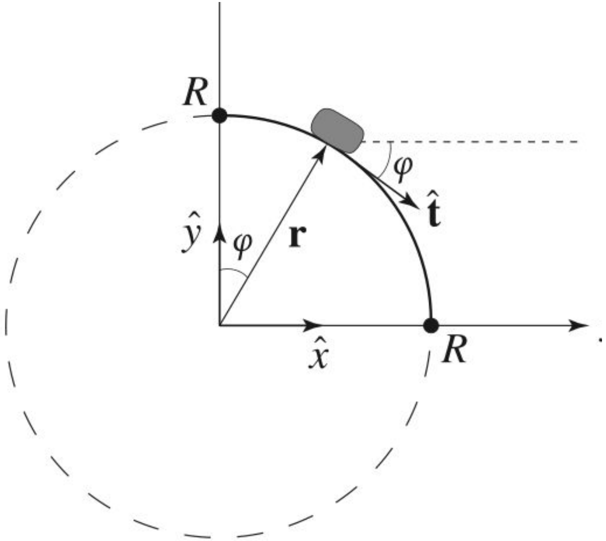
$$v^2(\varphi) = v_0^2\, e^{2\mu(\varphi-\varphi_0)} + 2ge^{2\mu\varphi} \times \int_{\varphi_0}^{\varphi} (\sin\varphi' - \mu\cos\varphi') \frac{e^{-2\mu\varphi'}}{\kappa(\varphi')}\, d\varphi',$$

Eq.2

$$H(\varphi) = \frac{1}{gR}\frac{\mathbf{g}\cdot\hat{\mathbf{n}}}{\kappa(\varphi)} = \frac{\cos\varphi}{R\kappa(\varphi)},$$

- By introducing the tangent angle, the particle velocity can be expressed as a function of the local geometry of the curve.
- Eq.1 describes how the velocity evolves alor the surface.
- Eq.2 defines the horizon function , which determines the condition for losing contact wit the surface.
- These equations simplify the analysis of mot on curved surfaces.

Departure from the circle



$$\mathbf{r}(\varphi) = (R\sin\varphi, R\cos\varphi), \text{ where } \varphi \in [0, \pi/2].$$

$$\kappa(\varphi) = 1/R,$$

$$H(\varphi) = \cos\varphi$$

Departure from the circle

<Condition for Leaving the Surface>

$$\frac{v^2(\varphi)}{gR} \le \cos\varphi,$$

$$\cos\varphi = \frac{2}{3} + \frac{v_0^2}{3gR},$$

<Special Case> $\mu = 0,$

$$v^2(\varphi) = v_0^2 + 2gR(1 - \cos\varphi),$$

Experimental process

Material lists

- Rectangular parallelepiped blocks
  Transparent hemisphere (21 cm in diameter)
- Digital camera
- Inclined plane (constructed from an L-shaped plastic folder)
- Measuring tape

Experimental procedure

1. The initial drop height of the object is measured.
2. The inclined plane is secured to the apex of the transparent hemisphere.
3. A camera is positioned perpendicularly to the ground to record the experiment.
4. The block is released from the inclined plane to slide downward.

Experimental procedure

5. The initial velocity at the apex of the hemisphere is calculated using the law of conservation of mechanical energy.

6. The detachment angle, at which the block separates from the surface of the hemisphere, is measured through video analysis.

7. The velocity at the exact moment of detachment is determined using the position and time coordinates of the block at an arbitrary point after separation.

Experimental procedure

8. The friction coefficient of the acrylic hemisphere is calculated by substituting the obtained values into the governing equations from previous literature.

9. The experimental error is determined by comparing the measured friction coefficient of the acrylic hemisphere with its theoretical value.

10. The potential sources of experimental error are analyzed and discussed

Experimental result

Our Experiment Video

Determining the Departure Point of the Block

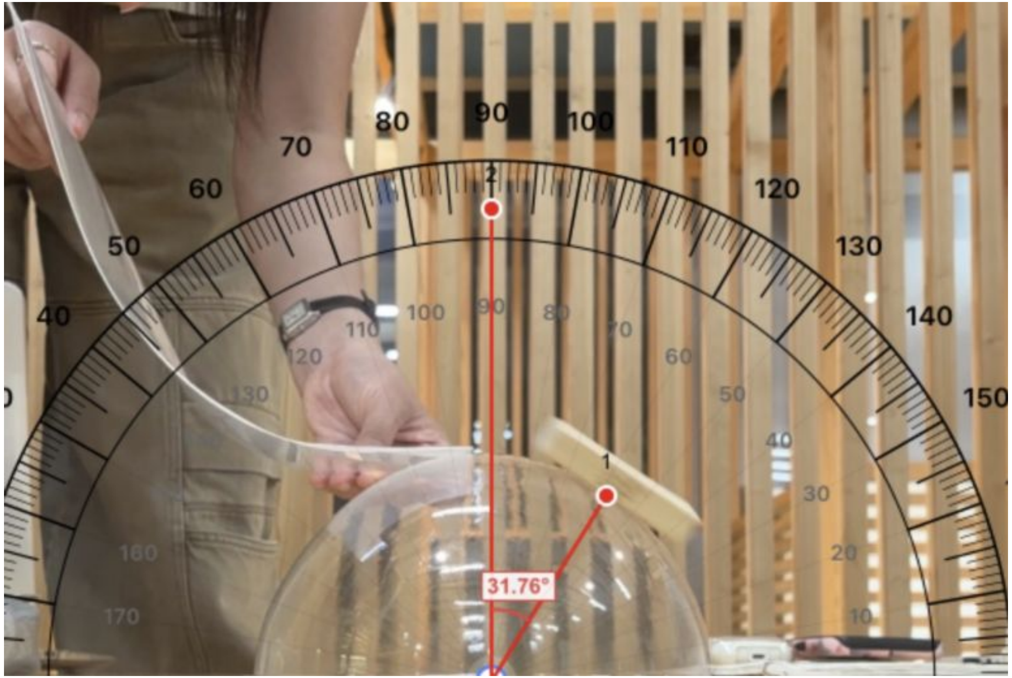


figure1:Point A

Measured Angle

$\varphi = 31.76° = 0.5543$ rad

Height at Point A

$x(A) = R\cos\varphi$

$= 0.101 \times \cos(31.76°) = 0.08588$ m

Horizontal Distance

$y(A) = R\sin\varphi$

$= 0.101 \times \sin(31.76°) = 0.05316$ m

Measured Time

$t(A) = 0.42$ s

Calculating the Velocity at the Departure Point

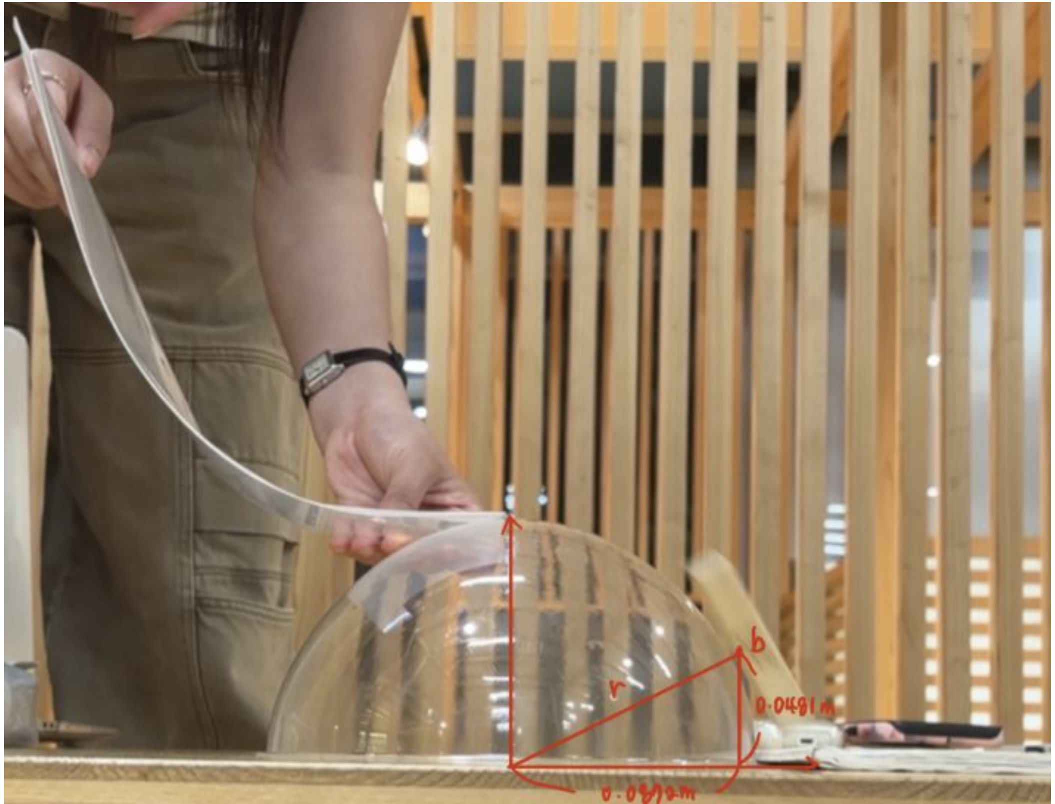


figure2:Point B

Calculating the Velocity at the Departure Point

Time Difference

$\Delta t = 0.47 - 0.42 = 0.05$ s

Horizontal Displacement

$\Delta x = 0.08722 - 0.05316$

$= 0.03406$ m

Vertical Displacement

$\Delta y = 0.08588 - 0.04810$

$= 0.03778$ m

Horizontal Velocity

$v_x = \Delta x/\Delta t = 0.03406/0.05 = 0.6812$ m/s

Vertical Velocity

$\Delta y = v_y\Delta t + \frac{1}{2}g\Delta t^2$

$0.03778 = v_y(0.05) + \frac{1}{2}(9.8)(0.05)^2$

$v_y = 0.5106$ m/s

Speed at Point A

$v_A = \sqrt{(v_x^2 + v_y^2)} = \sqrt{[(0.6812)^2 + (0.5106)^2]}$

$= 0.8513$ m/s

Calculating the Initial Velocity Using Energy Conservation

Given

- **Mass**
  $m = 0.0173$ kg
- **Initial Height**
  $h0 = 0.288$ m
- **Hemisphere Height**
  $htop = 0.101$ m

Initial Potential Energy

- **PEinitial**
  $= mgh$
  $= (0.0173)(9.8)(0.288)$
  $= 0.04883$ J

Potential Energy at Top

- **PEtop**
  $= (0.0173)(9.8)(0.101)$
  $= 0.01712$ J

Calculating the Initial Velocity Using Energy Conservation

Kinetic Energy

- **KE**
  $= 0.04883 - 0.01712 = 0.03170$ J
- **Initial Velocity**
  $KE = \frac{1}{2}mv^2$
  $v0 = \sqrt{(2KE/m)} = 1.914$ m/s

Applying the AJP Analytical Model

$$\frac{v^2(\varphi)}{gR} = \frac{v_0^2}{gR}\, e^{2\mu\varphi} + \frac{(2-4\mu^2)\left(e^{2\mu\varphi} - \cos\varphi\right) - 6\mu\sin\varphi}{1+4\mu^2}$$

figure3: Equation from the Paper

Known Values

$v(\varphi)$ = 0.8513 m/s

v_0 = 1.914 m/s

φ = 0.5543 rad

R = 0.101 m

g = 9.8 m/s²

Want to know : μ

Solving for the Friction Coefficient

Numerical Solution

- **Substituting all known values into the model:**
  $\mu = -1.17$

Comparison

- **Expected Acrylic Friction**
  $\mu \approx 0.83$
- **Experimental Result**
  $\mu = -1.17$

Conclusion

Why Did We Obtain a Negative Friction Coefficient?

Source 1

- **Overestimated Initial Velocity**
  Assumption: No energy loss before reaching the hemisphere
  Reality: Friction existed on the guiding track

=> v0 too large : 1.914 m/s

Why Did We Obtain a Negative Friction Coefficient?

Source 2

Underestimated Departure Velocity

- **Sources of error:**
  Air resistance
  Pixel scaling
  Motion blur
  Perspective distortion

=> vA too small : 0.851 m/s

Why Did We Obtain a Negative Friction Coefficient?

Result

Overestimated v0 + Underestimated vA

↓

Negative friction coefficient: $\mu = -1.17$

Conclusions

Achievements

✓ Identified the departure point experimentally

✓ Calculated departure velocity using projectile motion

✓ Calculated initial velocity using energy conservation

✓ Applied the AJP analytical model

✓ Investigated the discrepancy between theory and experiment

Conclusions

Final Takeaway

The most important outcome was not the numerical value of μ, but understanding how the intermediate mathematical operations connect theoretical equations to real physical motion.

Thank you